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Semiconductor device and its manufacturing method, memory and memory system

Abstract

Aspects of the disclosure provide a semiconductor device including a stack structure and a first conductive pillar. The stack structure can include a plurality of interlayer insulating layers and gate layers arranged alternately along a stack direction, and stack structure an array region and a staircase region adjoining the array region. The stack structure has a plurality of staircases in the staircase region arranged corresponding to the gate layers. Each of the gate layers can include a first gate part at the corresponding staircase and a second gate part connected to the first gate part, and a thickness of the first gate part along the stack direction is less than that of the second gate part along the stack direction. The first conductive pillar is arranged in the staircase region and penetrates through the stack structure along the stack direction, and the first conductive pillar is electrically connected with the first gate part.

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Background/Summary

INCORPORATION BY REFERENCE

(1) This present application claims the benefit of Chinese Patent Application No. 202211415775.6 filed on Nov. 11, 2022 which is incorporated herein by reference in its entirety.

BACKGROUND

Technical Field

(2) The present disclosure relates to the field of semiconductor technology, and in particular to a semiconductor device and its manufacturing method, a memory and a memory system.

Description of the Related Art

(3) With the development of planar flash memory, the production process of semiconductor has made great progress. But in recent years, the development of planar flash memory has encountered various challenges: physical limit, existing development technology limit and storage electron density limit. In order to solve the difficulties encountered by the planar flash memory and find a lower production cost per unit memory cell, three-dimensional (3D) flash memory structure came into being. In the structure of 3D NAND flash memory device, multi-layer gate layers and insulating layers that are stacked vertically and staggered are included, with channel holes being formed in the stack structure (or referred to as “stack”), memory cell strings being formed in the channel holes, the gate layers in the stack structure being used as the gate lines of the memory cells

in each layer, so as to realize the stacked 3D NAND flash memory device.

(4) At present, in the stack structure, it is necessary to etch a contact hole corresponding to each staircase and form a conductive pillar in the contact hole to achieve the purpose of inputting electrical signals to the gate layer at each staircase. However, in the process of etching the contact hole, it is easy to have bad phenomena such as etching through, which makes the conductive pillar short circuited between different gate layers and produces a crosstalk phenomenon.

SUMMARY

(5) Implementations of the present disclosure provides a semiconductor device and its manufacturing method, a memory and a memory system, which can avoid the occurrence of the phenomenon of short circuit between different gate layers caused by the etching accuracy of the first contact hole, and improve the yield of the semiconductor device.

(6) The implementation of the present disclosure provides a semiconductor device that can include a stack structure including a plurality of interlayer insulating layers and gate layers arranged alternately along a stack direction, wherein the stack structure includes an array region and a staircase region adjoining the array region. Further, the stack structure can have a plurality of staircases in the staircase region arranged corresponding to the gate layers, each of the gate layers includes a first gate part at the corresponding staircase and a second gate part connected to the first gate part, and a thickness of the first gate part along the stack direction is less than that of the second gate part along the stack direction. The semiconductor device that can also include a first conductive pillar arranged in the staircase region and penetrating through the stack structure along the stack direction, and the first conductive pillar is electrically connected with the first gate part.

(7) In an implementation of the present disclosure, the first conductive pillar penetrates through a plurality of the gate layers along the stack direction, and the semiconductor device further includes a conducting part between the first conductive pillar and the first gate part of one of the gate layers and arranged around the first conductive pillar and an insulating part between the first conductive pillar and the second gate parts of the other gate layers and arranged around the first conductive pillar. The first gate part can be arranged around the conducting part and in contact with the conducting part, the first conductive pillar can be electrically connected with the corresponding one first gate part through the conducting part, and the second gate part can be arranged around the insulating part.

(8) In an implementation of the present disclosure, a width of the conducting part along a first direction is less than that of the insulating part along the first direction, and the first direction is perpendicular to the stack direction.

(9) In an implementation of the present disclosure, the first gate part includes a conductive layer and a first dielectric layer at least between the conductive layer and the interlayer insulating layer, and the conductive layer is in contact with the conducting part, and a thickness of the conducting part along the stack direction is greater than that of the conductive layer along the stack direction.

(10) In an implementation of the present disclosure, the conducting part includes a conductive sub-part and a first adhesive buffer sub-part, the first adhesive buffer sub-part is between the first gate part and the conductive sub-part, and the conductive sub-part is between the first conductive pillar and the first adhesive buffer sub-part.

(11) In an implementation of the present disclosure, the first conductive pillar includes a conductive body and a second adhesive buffer sub-part covering the conductive body, and the second adhesive buffer sub-part is between the conductive sub-part and the conductive body.

(12) In an implementation of the present disclosure, the semiconductor device further includes a support pillar arranged in the staircase region, and the support pillar penetrates through the stack structure along the stack direction.

(13) In an implementation of the present disclosure, the semiconductor device includes a peripheral region on a side of the staircase region away from the array region, and a dielectric layer covering the stack structure and extending to the peripheral region. The semiconductor device can further

include a second conductive pillar arranged in the peripheral region and penetrating through the dielectric layer, and a material of the second conductive pillar is the same as that of the first conductive pillar.
(14) In an implementation of the present disclosure, the semiconductor device further includes a gate slit penetrating through the stack structure along the stack direction, and a semiconductor layer and an oxide layer filled in the gate slit, and the oxide layer covers a side wall of the semiconducting layer.
(15) According to the above objects, the implementation of the present application provides a manufacturing method of a semiconductor device. The manufacturing method of the semiconductor device includes the step of forming a stack structure which includes a plurality of interlayer insulating layers and a plurality of gate layers formed alternately along a stack direction, wherein the stack structure includes an array region and a staircase region adjoining the array region, a plurality of staircases corresponding to the gate layers are formed in the staircase region of the stack structure, each of the gate layers includes a first gate part at the corresponding staircase and a second gate part connected to the first gate part, and a thickness of the first gate part along the stack direction is less than that of the second gate part along the stack direction. The manufacturing method of the semiconductor device includes the step forming, in the staircase region, a first conductive pillar penetrating through the stack structure along the stack direction, the first conductive pillar being electrically connected with the first gate part.
(16) In an implementation of the present disclosure, the step of forming the stack structure further includes providing a substrate, and forming the plurality of interlayer insulating layers and gate sacrificial layers laminated along the stack direction on the substrate, wherein the plurality of interlayer insulating layers and the plurality of gate sacrificial layers form a plurality of initial staircases in the staircase region, and each of the initial staircases includes one of the gate sacrificial layers and one of the interlayer insulating layers, wherein the one of the gate sacrificial layers and the one of the interlayer insulating layers are in the staircase region and laminated. The step of forming the stack structure further includes thinning the gate sacrificial layer at each of the initial staircases, removing the gate sacrificial layer to form a plurality of slots, and forming the plurality of gate layers in the plurality of slots to obtain the stack structure.
(17) In an implementation of the present disclosure, after the step of thinning the gate sacrificial layer at each of the initial staircases, the manufacturing method further includes forming a first contact hole in the staircase region, the first contact hole penetrating through the stack structure along the stack direction, and forming an intermediate column in the first contact hole.
(18) In an implementation of the present disclosure, the step of forming the plurality of gate layers in the plurality of slots to obtain the stack structure further includes removing the intermediate column in the first contact hole, and making the first contact hole penetrate through the plurality of gate layers along the stack direction, etching the first gate part of one of the gate layers in the first contact hole to form a first groove, and etching the second gate part of the other gate layers in the first contact hole to form a second groove.
(19) In an implementation of the present disclosure, the step of forming, in the staircase region, a first conductive pillar penetrating through the stack structure along the stack direction further includes forming a conductive functional material layer in the first contact hole, wherein the conductive functional material layer is filled in the first groove and covers an inner wall of the second groove, and removing at least the conductive functional material layer outside the first groove to form a conducting part in the first groove, wherein the first gate part is around the conducting part and in contact with the conducting part. The step can further include forming an insulating part in the second groove, wherein the second gate part is around the insulating part, and forming the first conductive pillar corresponding to the first contact hole, wherein the first conductive pillar is electrically connected with the first gate part through the conducting part in the first contact hole.

(20) In an implementation of the present disclosure, the step of forming a conducting part in the first groove includes forming a conductive sub-part and a first adhesive buffer sub-part, wherein the first adhesive buffer sub-part is formed between the first gate part and the conductive sub-part, and the conductive sub-part is formed between the first conductive pillar and the first adhesive buffer sub-part.
(21) In an implementation of the present disclosure, the step of forming a plurality of the first conductive pillars corresponding to the respective first contact holes includes forming a conductive body and a second adhesive buffer sub-part covering the conductive body, wherein the second adhesive buffer sub-part is formed between the conductive sub-part and the conductive body.
(22) In an implementation of the present disclosure, the semiconductor device comprises a peripheral region on a side of the staircase region away from the array region and a dielectric layer formed at least in the peripheral region, the step of forming a first contact hole in the staircase region further includes using the same process to form the first contact hole, a second contact hole in the staircase region and form a third contact hole in the peripheral region, wherein the first contact hole and the second contact hole both penetrate through the stack structure along the stack direction, and the third contact hole penetrates through the dielectric layer along the stack direction.
(23) In an implementation of the present disclosure, the step of forming an intermediate column in the first contact hole further include forming a support pillar in the second contact hole, and the step of forming the first conductive pillar corresponding to the first contact hole further includes forming a second conductive pillar in the peripheral region, wherein the second conductive pillar is formed in the third contact hole.
(24) According to the above objects, the implementation of the present disclosure provides a memory, and the memory includes the semiconductor device.
(25) According to the above objects, the implementation of the present disclosure provides a memory system, and the memory system includes the memory and a controller which is coupled to the memory and used to control the memory to store data.
(26) According to the semiconductor device and its manufacturing method, a memory and a memory system provided by the implementation of the present disclosure, a plurality of first conductive pillars penetrating through the stack structure are arranged in the staircase region, and the first gate part of the gate layer at each staircase corresponds to at least one first conductive pillar and is electrically connected with the corresponding first conductive pillar. That is, the first conductive pillar in the present disclosure directly penetrates through the stack structure, and it is not necessary to accurately control the first conductive pillar to fall on each staircase. Therefore, the present disclosure can avoid the occurrence of the phenomenon of short circuit between different gate layers caused by the etching accuracy of the contact hole of the first conductive pillar, and improve the yield of semiconductor devices.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The technical solution and other beneficial effects of the present disclosure will be apparent through a detailed description of the specific implementations of the present disclosure in combination with the accompanying drawings.
(2) FIG. **1** is a schematic diagram of a cross-sectional structure of a semiconductor device in an exemplary implementation;
(3) FIG. **2** is a schematic diagram of another cross-sectional structure of a semiconductor device in an exemplary implementation;
(4) FIG. **3** is a schematic diagram of a cross-sectional structure of a part of region of a semiconductor device provided by an exemplary implementation of the present disclosure;

(5) FIG. **4** is a schematic diagram of a top view structure of a semiconductor device provided by an exemplary implementation of the present disclosure;
(6) FIG. **5** is a schematic diagram of a top view structure of a part of region of the semiconductor device provided by the implementation of the present disclosure;
(7) FIG. **6** is a schematic diagram of another top view structure of a part of region of a semiconductor device provided by an exemplary implementation of the present disclosure;
(8) FIG. **7** is a schematic diagram of another top view structure of a semiconductor device provided by an exemplary implementation of the present disclosure;
(9) FIG. **8** is a schematic diagram of another cross-sectional structure of a part of region of a semiconductor device provided by an exemplary implementation of the present disclosure;
(10) FIG. **9** is a schematic diagram of a partially enlarged cross-sectional structure of a semiconductor device in the staircase region provided by an exemplary implementation of the present disclosure;
(11) FIG. **10** is a flowchart of a manufacturing method of a semiconductor device provided by an exemplary implementation of the present disclosure;
(12) FIGS. **11** to **20** are schematic diagrams of structures of a manufacturing method of a semiconductor device provided by an exemplary implementation of the present disclosure;
(13) FIG. **21** is a schematic diagram of another cross-sectional structure of a part of region of a semiconductor device provided by an exemplary implementation of the present disclosure;
(14) FIG. **22** is a schematic diagram of another cross-sectional structure of a part of region of a semiconductor device provided by an exemplary implementation of the present disclosure;
(15) FIG. **23** is a schematic diagram of a structure of a memory provided by an exemplary implementation of the present disclosure; and
(16) FIG. **24** is a schematic diagram of a structure of a memory system provided by an exemplary implementation of the present disclosure.

DETAILED DESCRIPTION

(17) The technical solutions in the implementations of the present disclosure will be clearly and completely described below in combination with the accompanying drawings in the implementation of the present disclosure. Obviously, the described implementations are only a part of the implementations of the present disclosure, not all of them. Based on the implementations in the present disclosure, all the other implementations obtained by those skilled in the art without making inventive work fall within the protection scope of the present disclosure.
(18) The following description provides many different implementations or examples to implement different structures of the present disclosure. In order to simplify the description of the present disclosure, the components and arrangements of specific examples are described below. Of course, they are merely examples and are not intended to limit the present disclosure. In addition, reference numerals and/or reference letters may be repeated in different examples of the present disclosure. Such repetition is for the purpose of simplification and clarity, and does not in itself indicate the relationship between various discussed implementations and/or arrangements. In addition, the present disclosure provides examples of various specific processes and materials, but those skilled in the art can be aware of the application of other processes and/or the use of other materials.
(19) Referring to FIG. **1**, in the related art, the semiconductor device includes a stack structure, and the stack structure is formed by a plurality of gate conductive layers **1** and a plurality of gate insulating layers **2** alternated and laminated, the plurality of gate conductive layers **1** and gate insulating layers **2** form a plurality of staircases, and the semiconductor device includes a plurality of word line contacts **3**, so that each word line contact **3** falls on each staircase to realize the electrical connection between the word line contact **3** and the gate conductive layer **1** at the corresponding staircase. In the manufacturing process of semiconductor devices, it is usually necessary to pre-etch a contact hole, and then deposit a word line contact **3** in the contact hole. However, with the increase of the number of stack structures of the stack structure, the etching

depth of the contact hole is increasing, and the requirements for the etching process are also increasing. Due to process fluctuations and errors, the phenomenon of etching through is easy to occur in the etching process, leading to that the word line contact **3** connects the gate conductive layers **1** of different layers, resulting in signal crosstalk.
(20) Further to improve the structure of semiconductor devices in related technologies, referring to FIG. **2**, the gate conductive layer **4** at the staircase is thickened to avoid the occurrence of the phenomenon of etching through. However, in the process, the thickness of the gate sacrificial layer at the staircase will also increase so that after the subsequent removal of the gate sacrificial layer to form a slot, it is difficult for the conductive material to fill up the slot at the staircase and be removed in the subsequent etching process, resulting in that the subsequent word line contact **3** cannot be electrically connected with the gate conductive layer **4**. Therefore, in the related technology, it is difficult to make the word line contact **3** be connected with the gate conductive layer **1** (**4**) correspondingly while ensuring the yield.
(21) In order to solve the above problem, the implementation of the present disclosure provides a semiconductor device. Referring to FIG. **3**, the semiconductor device includes a stack structure and a first conductive pillar **20**.
(22) The stack structure includes a plurality of interlayer insulating layers **12** and gate layers **11** alternately arranged along the stack direction X. The stack structure includes an array region **101** and a staircase region **102** adjoining the array region **101**. The stack structure has, in the staircase region **102**, a plurality of staircases **10** arranged corresponding to the gate layers **11**. Each gate layer **11** includes a first gate part **110** at a corresponding staircase **10** and a second gate part **120** connected to the first gate part **110**, and the thickness of the first gate part **110** along the stack direction X is less than that of the second gate part **120** along the stack direction X. A first conductive pillar **20** is arranged in the staircase region **102** and penetrates through the stack structure along the stack direction X, and the first conductive pillar **20** is electrically connected with the first gate part **110**. That is, the implementation of the present disclosure realizes the electrical connection between the first conductive pillar **20** and the corresponding gate layer **11** by arranging the first conductive pillar **20** as penetrating through the stack structure and electrically connecting with the first gate part **110** at the staircase **10**, and thus does not need to accurately control the first conductive pillar **20** to fall on each staircase **10**. That is, the implementation of the present disclosure can avoid the occurrence of the phenomenon of short circuit between different gate layers **11** caused by the etching accuracy of the contact hole of the first conductive pillar **20**, and improve the yield of semiconductor devices.
(23) Specifically, referring to FIG. **4**, FIG. **5** and FIG. **6**, FIG. **4** shows a top view of a semiconductor device **700**, wherein the semiconductor device **700** includes one or more array regions **101**, and a staircase region **102** may be included between two adjacent array regions **101**.
(24) Further, FIG. **5** shows a schematic diagram of the partially enlarged top view structure of the above semiconductor device **700** at the array region **101** and the staircase region **102**, and FIG. **6** shows a schematic diagram of the partially enlarged top view structure of the above semiconductor device **700** at the array region **101**, wherein the array region **101** can be divided into a plurality of blocks by a gate slit **53**, and each block can be divided into a plurality of sub-regions by a top selection gate groove **61**, and a plurality of channel structures **54** may be included in each sub-region.
(25) In one implementation, there are a plurality of dummy channel structures **62** on the side of the staircase region **102** close to the array region **101**, which can be used to buffer stress and improve the reliability of the semiconductor device **700**. The staircase region **102** may include a plurality of staircases, and a first conductive pillar **20** is arranged corresponding to each staircase.
(26) In one implementation, support pillars can also be arranged at a plurality of staircases **10** to improve the stability of the stack structure, which will be described in detail in subsequent implementations.

(27) In addition, FIG. **7** shows a top view of another semiconductor device **700**, wherein in the semiconductor device **700**, the staircase region **102** may be on both sides of the array region **101**. It can be understood that the semiconductor device provided in the implementation of the present disclosure can be applied to any one of the structures in FIG. **4** and FIG. **7**.
(28) Further, the implementation of the present disclosure mainly improves the structure of the semiconductor device in the staircase region **102**. Referring to FIG. **3**, FIG. **8** and FIG. **9**, in the implementation of the present disclosure, the semiconductor device includes a stack structure, which includes an array region **101** and a staircase region **102** adjoining the array region **101**.
(29) Therein, the semiconductor device comprises a substrate (not shown in the FIG.) and a stack structure arranged on the substrate. The substrate includes, but is not limited to, Si substrate, Ge substrate, SiGe substrate, silicon on insulator (SOI) substrate or germanium on insulator (GOI) substrate, etc. and it can also be a silicon layer added after the formation of the stack structure. Therefore, the semiconductor layer can be used here to include these possible structures.
(30) The stack structure includes a plurality of gate layers **11** and interlayer insulating layers **12** arranged alternately along the stack direction X. The gate layer **11** includes a conductive layer and a first dielectric layer **113** covering a part of the surface of the conductive layer, wherein the first dielectric layer **113** may be between the conductive layer and the interlayer insulation layer **12** to achieve a better blocking effect on metal atoms in the conductive layer, and the first dielectric layer **113** is a dielectric layer with a high dielectric constant, such as a dielectric material with a dielectric constant K greater than 3.9. The conductive layer includes a metal sublayer **111** and a adhesive buffer sublayer **112** between the metal sublayer **111** and the first dielectric layer **113**.
(31) The semiconductor device further includes a peripheral region **103** on the side of the staircase region **102** away from the array region **101**, an isolation layer **41** covering a plurality of staircases **10**, and a dielectric layer **42** covering a plurality of staircases **10** and extending to the peripheral region **103**.
(32) In one implementation, the material of the interlayer insulation layer **12** may include silicon oxide material, the metal sublayer **111** may be tungsten, and may also include polysilicon or metal silicide material. For example, the metal silicide material may be provided as a silicide material including a metal selected from tungsten and titanium. The material of the adhesive buffer sublayer **112** may include titanium nitride, the material of the first dielectric layer **113** may include aluminum oxide or zirconium oxide, and the materials of the isolation layer **41** and the dielectric layer **42** may include a silicon oxide material.
(33) Further, the semiconductor device also includes a gate slit **53** penetrating through the stack structure and a channel structure **54** in the array region **101**. It can be understood that the gate slit **53** can be filled with a semiconductor layer and an oxide layer covering the side wall of the semiconductor layer, wherein the material of the semiconductor layer can include a polysilicon material, and the material of the oxide layer can include a silicon oxide material.
(34) The channel structure **54** is formed in the channel hole, and can include a charge storage layer, a tunneling layer and a channel layer from the outside to the inside, and filling materials can be further deposited in the channel hole to completely or partially fill the channel hole to form the channel structure **54**.
(35) In one implementation, the material of the tunneling layer may include silicon oxide, the material of the charge storage layer may include silicon nitride or silicon oxynitride, the material of the channel layer may include polysilicon, and the filling material may include silicon oxide or other dielectric materials.
(36) In one implementation, the gate layer **11** includes a conductive layer, but does not include the first dielectric layer **113** which can be selectively arranged in the channel hole.
(37) In addition, a plurality of staircases **10** corresponding to the plurality of gate layers **11** one-to-one can be formed in the staircase region **102** of the stack structure. In the implementation of the present disclosure, each gate layer **11** includes a first gate part **110** at the corresponding staircase **10**

and a second gate part **120** connected to the first gate part **110** and extending into the array region **101**. Therein, the thickness H**1** of the first gate part **110** along the stack direction X is less than the thickness H**2** of the second gate part **120** along the stack direction X.
(38) In one implementation, the total thickness of the metal sublayer **111**, the adhesive buffer sublayer **112**, and the first dielectric layer **113** in the first gate part **110** is less than the total thickness of the metal sublayer **111**, the adhesive buffer sublayer **112**, and the first dielectric layer **113** in the second gate part **120**.
(39) Further, the semiconductor device provided by the implementation of the present disclosure further includes a plurality of first conductive pillars **20** arranged in the staircase region **102**, each of the first conductive pillars **20** penetrates through the stack structure along the stack direction X, that is, each of the first conductive pillars **20** penetrates through a plurality of gate layers **11** along the stack direction X, wherein the plurality of first conductive pillars **20** correspond, one-to-one, to the plurality of first gate parts **110** at the plurality of staircases **10** and are electrically connected with them. Correspondingly, the first conductive pillar **20** is also insulated from at least one second gate part **120** on one side of the first gate part **110** along the stack direction X, so that a plurality of first conductive pillars **20** can be electrically connected to a plurality of gate layers **11**. By thinning the thickness of the first gate part **110**, the implementation of the present disclosure can ensure that the first gate part **110** at the staircase **10** can be fully filled during the formation process, reducing the risk of the occurrence of voids in the first gate part **110** and further reducing the risk of poor conducting between the first conductive pillar **20** and the first gate part **110**.
(40) It should be noted that the first conductive pillar **20** penetrates through the dielectric layer **42**, the isolation layer **41** and the stack structure along the stack direction X.
(41) Specifically, the semiconductor device includes a conducting part **31** between the first conductive pillar **20** and the first gate part **110** and arranged around the first conductive pillar **20**, and an insulating part **32** between the first conductive pillar **20** and the second gate part **120** and arranged around the first conductive pillar **20**, wherein the first gate part **110** is arranged around the conducting part **31** and is in contact with the conducting part **31**, and the second gate part **120** is arranged around the insulating part **32** and is insulated from the first conductive pillar **20** through the insulating part **32**.
(42) Therein, the plurality of first conductive pillars **20** include one first conductive pillar **20** penetrating through one gate layer **11** along the stack direction X, and a conducting part **31** is arranged between the first conductive pillar **20** and the first gate part **110**. The other first conductive pillars **20** penetrate at least two gate layers **11** along the stack direction X, and a conducting part **31** is arranged between the other first conductive pillars **20** and the first gate part **110** of one of the gate layers **11**, and an insulating part **32** is arranged between the other first conductive pillars **20** and the second gate part **120** of the other gate layers **11**.
(43) It can be understood that the plurality of first conductive pillars **20** include a plurality of first conductive pillars **20** penetrating through at least two gate layers **11** along the stack direction X and one first conductive pillar **20** penetrating through one gate layer **11** along the stack direction X, wherein among the plurality of first conductive pillars **20** penetrating through at least two gate layers **11** along the stack direction X, each first conductive pillar **20** is electrically connected with one gate layer **11** through the conducting part **31** and is insulated from the other gate layers **11** through the insulating part **32**. In the one first conductive pillar **20** penetrating through one gate layer **11** along the stack direction X, the first conductive pillar **20** is electrically connected with the one gate layer **11** which it penetrates through via the conducting part **31**. In the implementation of the present disclosure, a plurality of first conductive pillars **20** can directly penetrate through the stack structure, thereby reducing the demand for etching accuracy, reducing the process difficulty and improving the yield.
(44) In one implementation, the conducting part **31** includes a conductive sub-part **311** and a first adhesive buffer sub-part **312**, the first adhesive buffer sub-part **312** is between the first gate part

110 and the conductive sub-part **311**, and the conductive sub-part **311** is between the first conductive pillar **20** and the first adhesive buffer sub-part **312**. Therein, the conductive layer is exposed on the surface of the first gate part **110** on the side close to the conducting part **31**, that is, the metal sublayer **111** is exposed, and the metal sublayer **111** is enabled to be in direct contact with the conducting part **31**, that is, in direct contact with the first adhesive buffer sub-part **312**, so that the first gate part **110** can be electrically connected with the conducting part **31**. The conductive layer is exposed on the surface of the second gate part **120** on the side close to the insulating part **32**, that is, the metal sublayer **111** is exposed, and the metal sublayer **111** is enabled to be in contact with the insulating part **32**, and the insulating part **32** completely covers the surface of the metal sublayer **111** on the side close to the first conductive pillar **20**, so that the second gate part **120** can be arranged separately from the first conductive pillar **20** through the insulating part **32**.

(45) In one implementation, the first conductive pillar **20** includes a conductive body **21** and a second adhesive buffer sub-part **22** covering the conductive body **21**, and the second adhesive buffer sub-part **22** is between the conductive sub-part **311** and the conductive body **21**, so that the first conductive pillar **20** is electrically connected with the conducting part **31**, and thus the first conductive pillar **20** can be electrically connected with the first gate part **110** through the conducting part **31**.

(46) Therein, the material of the conductive sub-part **311** and the material of the conductive body **21** may include tungsten, and the material of the first adhesive buffer sub-part **312** and the material of the second adhesive buffer sub-part **22** may include titanium nitride.

(47) In one implementation, the thickness of the conducting part **31** along the stack direction X is equal to that of the first gate part **110** along the stack direction X, and thus the thickness of the conducting part **31** along the stack direction X is greater than that of the conductive layer in the first gate part **110** along the stack direction X, while the thickness of the insulating part **32** along the stack direction X is equal to that of the second gate part **120** along the stack direction X.

(48) Since the thickness of the first gate part **110** along the stack direction X is greater than that of the second gate part **120** along the stack direction X, the thickness of the conducting part **31** along the stack direction X is less than that of the insulating part **32** along the stack direction X. It should be noted that, by reducing the thickness of the first gate part **110** to be less than that of the second gate part **120**, the implementation of the present disclosure can reduce the thickness of the groove that needs to be filled when the conducting part **31** is formed, so that the conducting part **31** can fill up the entire groove, thereby improving the success rate of the electrical connection between the conducting part **31** and the first conductive pillar **20**.

(49) Further, the width W**1** of the conducting part **31** along the first direction Y is smaller than the width W**2** of the insulating part **32** along the first direction Y, and the first direction Y is perpendicular to the stack direction X.

(50) In the implementation of the present disclosure, the semiconductor device further comprises: a support pillar **52** arranged in the staircase region **102** and penetrating through the dielectric layer **42**, the isolation layer **41** and the stack structure; an oxide layer **43** covering the side wall of the support pillar **52**, the upper surface of the dielectric layer **42** and the upper surface of the stack structure; and a cover layer **44** covering the upper surface of the oxide layer **43**. Because in the implementation of the present disclosure, a plurality of first conductive pillars **20** all penetrate through the stack structure, in order to improve the stability of the stack structure, in the implementation of the present disclosure, the support pillar **52** is arranged in the staircase region **102** to improve the stability of the stack structure and the reliability of the semiconductor device.

(51) In one implementation, each staircase **10** is correspondingly provided with at least one support pillar **52**.

(52) In one implementation, the material of the support pillar **52** may include a polysilicon material, and the materials of the oxide layer **43** and the cover layer **44** may include a silicon oxide material.

(53) Further, the semiconductor device also includes a second conductive pillar **51** arranged in the peripheral region **103** and penetrating through the dielectric layer **42** and the isolation layer **41** along the stack direction X. It should be noted that the material of the second conductive pillar **51** can be the same as that of the first conductive pillar **20**, that is, the second conductive pillar **51** can include a conductive column made of tungsten and a third adhesive buffer sub-part covering the side wall of the conductive column and made of titanium nitride. Thus, the first conductive pillar **20** and the second conductive pillar **51** can be prepared in the same process to simplify the process, reduce the process time and reduce the process cost.
(54) Following the above, in the implementation of the present disclosure, a plurality of first conductive pillars **20** penetrating through the stack structure are arranged in the staircase region **102**, and each first conductive pillar **20** is electrically connected with the first gate part **110** of the gate layer **11** at one staircase **10** correspondingly, so that the plurality of first conductive pillars **20** correspond to the plurality of gate layers **11** one-to-one and are electrically connected with them. That is, the first conductive pillar **20** in the implementation of the present disclosure directly penetrates through the stack structure, and it is not necessary to accurately control the first conductive pillar **20** to fall on each staircase **10**. Therefore, the implementation of the present disclosure can avoid the occurrence of the phenomenon of short circuit between different gate layers **11** caused by the etching accuracy of the contact hole of the first conductive pillar **20**, and improve the yield of semiconductor devices.
(55) In addition, the implementation of the present disclosure also provides a manufacturing method of a semiconductor device. Referring FIG. **3**, FIG. **8**, FIG. **9** and FIG. **10**, the manufacturing method of the semiconductor device includes:
(56) Forming a stack structure. The stack structure includes a plurality of interlayer insulating layers **12** and a plurality of gate layers **11** formed alternately along the stack direction X. The stack structure includes an array region **101** and a staircase region **102** adjoining the array region **101**. A plurality of staircases **10** corresponding to the gate layers **11** are formed in the staircase region **102** of the stack structure. Each gate layer **11** includes a first gate part **110** at the corresponding staircase **10** and a second gate part **120** connected to the first gate part **110**, and the thickness of the first gate part **110** along the stack direction X is less than the thickness of the second gate part **120** along the stack direction.
(57) Forming, in the staircase region **102**, a first conductive pillar **20** penetrating through the stack structure along the stack direction X, the first conductive pillar **20** being electrically connected with the first gate part **110**.
(58) That is, the implementation of the present disclosure realizes the electrical connection between the first conductive pillar **20** and the corresponding gate layer **11** by forming the first conductive pillar **20** as penetrating through the stack structure and electrically connecting the first gate part **110** at the staircase **10**, thus it is not necessary to accurately control the first conductive pillar **20** to fall on each staircase **10**, that is, the implementation of the present disclosure can avoid the occurrence of the phenomenon of short circuit between different gate layers **11** caused by the etching accuracy of the contact hole of the first conductive pillar **20**, and improve the yield of the semiconductor device. By thinning the thickness of the first gate part **110**, the implementation of the present disclosure can ensure that the first gate part **110** at the staircase **10** can be fully filled during the formation process, reducing the risk of the occurrence of voids in the first gate part **110** and further reducing the risk of poor conducting between the first conductive pillar **20** and the first gate part **110**.
(59) Specifically, referring to FIG. **3**, FIG. **8**, FIG. **9**, FIG. **10**, FIG. **11** to FIG. **20**, the manufacturing method of the semiconductor device includes:
(60) S**10**: A stack structure including a plurality of interlayer insulating layers **12** and a plurality of gate layers **11** formed alternately along the stack direction X is formed. The stack structure includes an array region **101** and a staircase region **102** adjoining the array region **101**. A plurality of

staircases **10** corresponding to the gate layers **11** are formed in the staircase region **102** of the stack structure. Each gate layer **11** includes a first gate part **110** at the corresponding staircase **10** and a second gate part **120** connected to the first gate part **110**, and the thickness of the first gate part **110** along the stack direction X is less than that of the second gate part **120** along the stack direction.
(61) In step S**10**, a substrate (not shown in the FIGS.) is first provided, and the substrate includes but not limited to Si substrate, Ge substrate, SiGe substrate, silicon on insulator (SOI) substrate or germanium on insulator (GOI) substrate, etc.
(62) A plurality of interlayer insulating layers **12** and gate sacrificial layers **13** laminated along the stack direction X are formed on the substrate.
(63) Then, a channel structure **54** is formed in the stack structure corresponding to the array region **101**, wherein the channel structure **54** is formed in the channel hole and can include a charge storage layer, a tunneling layer and a channel layer from the outside to the inside, and filling materials can be further deposited in the channel hole to completely or partially fill the channel hole to form the channel structure **54**.
(64) In one implementation, the material of the tunneling layer may include silicon oxide, the material of the charge storage layer may include silicon nitride or silicon oxynitride, the material of the channel layer may include polysilicon, and the filling material may include silicon oxide or other dielectric materials.
(65) Next, the parts of the plurality of interlayer insulating layers **12** and the plurality of gate sacrificial layers **13** in the staircase region **102** are etched to form a plurality of initial staircases **1100** in the staircase region **102**. Each initial staircase **1100** includes one gate sacrificial layer **13** and one interlayer insulating layer **12** in the staircase region **102** and laminated. Therein, the material of the interlayer insulation layer **12** includes silicon oxide material, and the material of the gate sacrificial layer **13** includes silicon nitride material, as shown in FIG. **11**.
(66) Then, the gate sacrificial layer **13** at each initial staircase **1100** is thinned so that the thickness of the gate sacrificial layer **13** at the initial staircase **1100** becomes smaller and less than the thickness of the gate sacrificial layer **13** outside the initial staircase **1100**, as shown in FIG. **12**.
(67) Next, the isolation layer **41** is deposited to cover the plurality of initial staircases **1100**, and the dielectric layer **42** is deposited to cover the isolation layer **41** and extend to the peripheral region **103** of the semiconductor device.
(68) In one implementation, the material of the isolation layer **41** and the material of the dielectric layer **42** may include a silicon oxide material.
(69) Using the same process, a plurality of first contact holes **401**, a plurality of second contact holes **402** are formed in the staircase region **102**, and a plurality of third contact holes **403** and a gate slit structure **404** are formed in the peripheral region **103**. The first contact holes **401**, the second contact holes **402** penetrate through the dielectric layer **42**, the isolation layer **41**, the plurality of interlayer insulation layers **12** and the plurality of gate sacrificial layers **13** along the stack direction X, the third contact holes **403** penetrates through the dielectric layer **42** along the stack direction X, and the gate slit structure **404** penetrates through the plurality of interlayer insulating layers **12** and the plurality of gate sacrificial layers **13** along the stack direction X.
(70) It should be noted that the first contact hole **401** and the second contact hole **402** can directly penetrate through the stack structure, and thus reduce the demand for etching accuracy, reduce the process difficulty and improve the yield.
(71) Then, an intermediate column **210** is formed in the first contact hole **401**, the third contact hole **403**, and the gate slit structure **404**, and a support pillar **52** is formed in the second contact hole **402**. Therein, the material of the intermediate column **210** and the material of the support pillar **52** may include polysilicon material, and when depositing the intermediate column **210** and the support pillar **52**, an oxide layer **43** may be formed on the inner walls of the first contact hole **401**, the second contact hole **402**, the third contact hole **403**, and the gate slit structure **404**, and on the upper surface of the dielectric layer **42**, and the oxide layer **43** may be a silicon oxide layer, as

shown in FIG. **13**.
(72) Next, the gate sacrificial layer **13** is removed to form a plurality of slots.
(73) Further, a plurality of gate layers **11** are formed in the plurality of slots to obtain a stack structure. Therein, the step of forming the gate layer **11** includes forming a first dielectric layer **113** and a conductive layer, and the first dielectric layer **113** covers a part of the surface of the conductive layer, wherein the first dielectric layer **113** can be between the conductive layer and the interlayer insulation layer **12**, so as to achieve a better blocking effect on the metal atoms in the conductive layer. The conductive layer includes a metal sublayer **111** and a adhesive buffer sublayer **112** between the metal sublayer **111** and the interlayer insulation layer **12**.
(74) Therein, a plurality of staircases **10** corresponding to a plurality of gate layers **11** are formed in the staircase region **102** of the stack structure. It can be understood that the plurality of formed staircases **10** can correspond to the plurality of initial staircases **1100** one-to-one. The gate layer **11** includes a first gate part **110** at the corresponding staircase **10** and a second gate part **120** connected to the first gate part **110**. Since the part of the gate sacrificial layer **13** at the initial staircase **1100** is thinned in the implementation of the present disclosure, after the gate layer **11** is formed, the thickness of the first gate part **110** at the staircase **10** along the stack direction X is less than the thickness of the second gate part **120** along the stack direction X. That is, by thinning the thickness of the first gate part **110**, the implementation of the present disclosure can ensure that the first gate part **110** at the staircase **10** can be fully filled during the formation process, reducing the risk of the occurrence of voids in the first gate part **110** and further reducing the risk of poor conducting between the first conductive pillar **20** and the first gate part **110**.
(75) In one implementation, the metal sublayer **111** may be tungsten, and may also include polysilicon or a metal silicide material. For example, the metal silicide material may be provided as a silicide material including a metal selected from tungsten and titanium, the material of bonding the buffer sublayer **112** may include titanium nitride, and the material of the first dielectric layer **113** may include aluminum oxide or zirconium oxide.
(76) Next, the oxide layer **43** and the intermediate column **210** in the gate slit structure **404**, as well as a part of the film layers of the plurality of gate layers **11** are removed. For example, the first dielectric layer **113**, the adhesive buffer sublayer **112** and a part of the metal sublayer **111** of the gate layer **11** on the side close to the gate slit structure **404** are removed to expose the metal sublayer **111** on the side close to the gate slit structure **404**.
(77) Then, an insulating oxide layer **532** and a filler **531** are deposited in the gate slit structure **404** to form a gate slit **53**, and the insulating oxide layer **532** covers the side wall of the filler **531**. A cover layer **44** is deposited on the oxide layer **43** and the gate slit **53**, as shown in FIG. **14**.
(78) In one implementation, the material of the cover layer **44** may include a silicon oxide material, the material of the insulating oxide layer **532** may include a silicon oxide material, and the material of the filler **531** may include a polysilicon material.
(79) Further, the intermediate column **210** in the first contact hole **401** and the intermediate column **210** in the third contact hole **403** are removed, as shown in FIG. **15**.
(80) The plurality of first contact holes **401** include one first contact hole **401** penetrating through one gate layer **11** along the stack direction X, and the first contact hole **401** is at the outermost periphery of the staircase region **102**. The first gate part **110** is etched in the first contact hole **401** to form a first groove **4011**. The other first contact holes **401** penetrate through at least two gate layers **11** along the stack direction X, and then the first gate part **110** of one of the gate layers **11** is etched in the other first contact holes **401** to form a first groove **4011**, and the second gate parts **120** of the other gate layers **11** are etched to form second grooves **4012**, and the plurality of first grooves **4011** correspond to the first gate parts **110** of the plurality of gate layers **11** one-to-one, as shown in FIG. **16**.
(81) Therein, the inner wall of the first groove **4011** exposes the metal sublayer **111** in the first gate part **110**, and the inner wall of the second groove **4012** exposes the metal sublayer **111** in the

second gate part **120**.

(82) S**20**: A plurality of first conductive pillars **20** penetrating through the stack structure along the stack direction X are formed in the staircase region **102**, and the first conductive pillars **20** are electrically connected with the first gate parts **110**.

(83) In step S**20**, a conductive functional material layer **33** is formed in a plurality of first contact holes **401**, and the conductive functional material layer **33** is filled in the first groove **4011** and covers the inner wall of the second groove **4012**, and the conductive functional material layer **33** includes a metal material layer **331** and a dielectric layer **332**, as shown in FIG. **17**.

(84) It should be noted that since the thickness of the first gate part **110** is less than that of the second gate part **120**, the size of the first groove **4011** along the stack direction X is less than that of the second groove **4012** along the stack direction X, and thus the conductive functional material layer **33** can be completely filled in the first groove **4011** and cover along the inner wall of the second groove **4012**, and cannot completely fill the second groove **4012**.

(85) In one implementation, the material of the metal material layer **331** may include tungsten, and the material of the dielectric layer **332** may include titanium nitride.

(86) Then, at least the conductive functional material layer **33** outside the first groove **4011** is removed to form a conducting part **31** in the first groove **4011**, and the first gate part **110** is around and in contact with the conducting part **31**, as shown in FIG. **18**.

(87) Therein, the conductive functional material layer **33** outside the first groove **4011** is removed, and a part of the conductive functional material layer **33** in the first groove **4011** is removed, to form a conducting part **31** in the first groove **4011**. The conducting part **31** includes a conductive sub-part **311** filled in the first groove **4011**, and a first adhesive buffer sub-part **312** covering the inner wall of the first groove **4011** and being between the conductive sub-part **311** and the inner wall of the first groove **4011**, and the conductive sub-part **311** is exposed to the inner wall of the first contact hole **401**.

(88) It can be understood that in the process of removing the conductive functional material layer **33**, it is necessary to control the etching amount in the first groove **4011** along the first direction Y to be less than that in the second groove **4012** along the first direction Y, so as to ensure that the conducting part **31** can be formed in the first groove **4011**.

(89) Next, an insulating layer **34** is deposited on the upper surfaces of the first contact hole **401**, the third contact hole **403**, and the cover layer **44**, and the insulating layer **34** continuously covers the first groove **4011** and the second groove **4012**, and is filled in the second groove **4012** to cover the metal sublayer **111** of the second gate part **120** exposed at the second groove **4012**, as shown in FIG. **19**. And the material of the insulating layer **34** may include a silicon oxide material.

(90) Then, the insulating layer **34** outside the second groove **4012** is removed to form an insulating part **32** in the second groove **4012**, and the insulating part **32** is filled in the second groove **4012** and covers the exposed metal sublayer **111** of the second gate part **120**, as shown in FIG. **20**.

(91) Further, a plurality of first conductive pillars **20** are formed corresponding to each first contact hole **401**, and a second conductive pillar **51** is formed in the third contact hole **403**, and one first conductive pillar **20** is electrically connected with the first gate part **110** through the conducting part **31** in one corresponding first contact hole **401**, as shown in FIG. **8**.

(92) Therein, a second adhesive buffer sub-part **22** and a conductive body **21** are formed in turn in the first contact hole **401** to form the first conductive pillar **20**; and a third adhesive buffer sub-par **512** and a conductive column **511** are formed in turn in the third contact hole **403**. The second adhesive buffer sub-part **22** covers the side wall of the conductive body **21** and is between the conductive sub-part **311** and the conductive body **21**, and the third adhesive buffer sub-part **512** covers the side wall of the conductive column **511**.

(93) In one implementation, the material of the second adhesive buffer sub-part **22** and the material of the third adhesive buffer sub-part **512** may include titanium nitride, and the material of the conductive body **21** and the material of the conductive column **511** may include tungsten.

(94) It should be noted that in the manufacturing method of the above semiconductor device, when the gate sacrificial layer **13** at the initial staircase **1100** is thinned using dry etching, the gate sacrificial layer **13** and the interlayer insulation layer **12** at the initial staircase **1100** can also be etched away at the same time, so that the side wall between the two adjacent initial staircases **1100** is a flush surface, as shown in FIG. **8** and FIG. **11**, and when the gate sacrificial layer **13** at the initial staircase **1100** is thinned using wet etching, only a part of the gate sacrificial layer **13** is removed, so that the interlayer insulation layer **12** protrudes from the side wall between the two adjacent initial staircases **1100**, and all the film layers formed in the subsequent process conduct covering conformally to the protruding part of the interlayer insulation layer **12**, as shown in FIG. **21** and FIG. **22**.
(95) Following the above, in the implementation of the present disclosure, a plurality of first conductive pillars **20** penetrating through the stack structure are arranged in the staircase region **102**, and each first conductive pillar **20** is correspondingly electrically connected with the first gate part **110** of the gate layer **11** at one staircase **10**, so that the plurality of first conductive pillars **20** can correspond to the plurality of gate layers **11** one-to-one and be electrically connected with them. That is, the first conductive pillar **20** in the implementation of the present disclosure directly penetrates through the stack structure, and it is not necessary to accurately control the first conductive pillar **20** to fall on each staircase **10**. Therefore, the implementation of the present disclosure can avoid the occurrence of the phenomenon of short circuit between different gate layers **11** caused by the etching accuracy of the contact hole of the first conductive pillar **20**, and improve the yield of semiconductor devices.
(96) Accordingly, referring to FIG. **23**, the implementation of the present disclosure also provides a memory **80**, which includes a memory chip **81** and a peripheral circuit **82**. The peripheral circuit **82** is electrically connected with the memory chip **81**, and the peripheral circuit **82** can store data in or read data from the memory chip **81**. Therein, the memory chip **81** and/or the peripheral circuit **82** may include a semiconductor device in any of the above implementations.
(97) Specifically, the above memory **80** may be specifically a three-dimensional memory (e.g., a 3D NAND memory).
(98) It can be understood that the memory provided by the implementation of the present disclosure has the same beneficial effect as the semiconductor device provided by the implementation of the present disclosure since the semiconductor device is arranged therein.
(99) In addition, the implementation of the present disclosure also provides a memory system. Referring to FIG. **24**, the memory system **91** includes a controller **911** and a memory **80** in any of the above implementations. The controller **911** is coupled to the memory **80** and is used to control the memory **80** to store data.
(100) Specifically, the controller **911** may control the memory **80** through the channel CH, and the memory **80** may perform operations based on the control of the controller **911** in response to a request from the host **92**. The memory **80** may receive a command CMD and an address ADDR from the controller **911** through the channel CH and access a region selected from a memory cell array in response to the address. In other words, the memory **80** may perform an internal operation corresponding to the command on the region selected by the address.
(101) In some implementations, the memory system **91** may be implemented as a universal flash memory (UFS) device, a solid state disk (SSD), a multimedia card in the form of MMC, eMMC, RS-MMC, and micro MMC, a secure digital card in the form of SD, mini SD, and micro SD, a memory device of the personal computer memory card International Association (PCMCIA) card type, a memory device of the peripheral component interconnect (PCI) type, a memory device of the high-speed PCI (PCI-E) type, and a compact flash (CF) card, smart media card or memory stick, etc.
(102) Specifically, the above memory system **91** can be used in terminal products such as computers, televisions, set-top boxes, and vehicles.

(103) Further, the implementation of the present disclosure also provides an electronic device, which includes the above memory system **91** provided by the implementation of the present disclosure. Specifically, the electronic device can be any device that can store data, such as mobile phones, desktop computers, tablets, notebooks, servers, vehicular devices, wearable devices, mobile power supplies, etc.
(104) The electronic device provided by the implementation of the present disclosure has the same beneficial effect as the above memory system provided by the implementation of the present disclosure as the memory system is arranged therein.
(105) In the above implementations, the description of each implementation has its own emphasis. For the part not detailed in one implementation, referring to the relevant description of other implementations.
(106) The above describes in detail a semiconductor device and its manufacturing method, a memory and a memory system provided by the implementations of the present disclosure. A specific example is applied herein to explain the principle and implementation of the present disclosure. The description of the above implementation is only used to help understand the technical solutions and their core ideas of the present disclosure. Those ordinary skill in the art should understand that they can still modify the technical solutions recited in the above implementations, or make equivalent replacement for some of the technical features. These modifications or replacements do not make the essence of the corresponding technical solutions depart from the scope of the technical solutions of the implementations of the present disclosure.

Claims

1. A semiconductor device, comprising: a stack structure including a plurality of interlayer insulating layers and gate layers arranged alternately along a stack direction, wherein the stack structure includes an array region and a staircase region adjoining the array region, the stack structure has a plurality of staircases in the staircase region arranged corresponding to the gate layers, each of the gate layers includes a first gate part at the corresponding staircase and a second gate part connected to the first gate part, and a thickness of the first gate part along the stack direction is less than that of the second gate part along the stack direction; and a first conductive pillar arranged in the staircase region and penetrating through the stack structure along the stack direction, wherein the first conductive pillar is electrically connected with the first gate part.
2. The semiconductor device according to claim 1, wherein the first conductive pillar penetrates through a plurality of the gate layers along the stack direction, and the semiconductor device further comprises: a conducting part between the first conductive pillar and the first gate part of one of the gate layers and arranged around the first conductive pillar; and an insulating part between the first conductive pillar and the second gate parts of the other gate layers and arranged around the first conductive pillar, wherein the first gate part is arranged around the conducting part and in contact with the conducting part, the first conductive pillar is electrically connected with the corresponding one first gate part through the conducting part, and the second gate part is arranged around the insulating part.
3. The semiconductor device according to claim 2, wherein a width of the conducting part along a first direction is less than that of the insulating part along the first direction, and the first direction is perpendicular to the stack direction.
4. The semiconductor device according to claim 2, wherein the first gate part further comprises a conductive layer and a first dielectric layer at least between the conductive layer and the interlayer insulating layer, and the conductive layer is in contact with the conducting part, and a thickness of the conducting part along the stack direction is greater than that of the conductive layer along the stack direction.
5. The semiconductor device according to claim 2, wherein the conducting part further comprises a

conductive sub-part and a first adhesive buffer sub-part, the first adhesive buffer sub-part is between the first gate part and the conductive sub-part, and the conductive sub-part is between the first conductive pillar and the first adhesive buffer sub-part.
6. The semiconductor device according to claim 5, wherein the first conductive pillar further comprises a conductive body and a second adhesive buffer sub-part covering the conductive body, and the second adhesive buffer sub-part is between the conductive sub-part and the conductive body.
7. The semiconductor device according to claim 1, wherein the semiconductor device further comprises a support pillar arranged in the staircase region, and the support pillar penetrates through the stack structure along the stack direction.
8. The semiconductor device according to claim 1, wherein the semiconductor device further comprises: a peripheral region on a side of the staircase region away from the array region, and a dielectric layer covering the stack structure and extending to the peripheral region; and a second conductive pillar arranged in the peripheral region and penetrating through the dielectric layer, and a material of the second conductive pillar is the same as that of the first conductive pillar.
9. The semiconductor device according to claim 1, wherein the semiconductor device further comprises a gate slit penetrating through the stack structure along the stack direction, and a semiconductor layer and an oxide layer filled in the gate slit, and the oxide layer covers a side wall of the semiconducting layer.
10. A manufacturing method of a semiconductor device comprising the steps of: forming a stack structure which includes a plurality of interlayer insulating layers and a plurality of gate layers formed alternately along a stack direction, wherein the stack structure includes an array region and a staircase region adjoining the array region, a plurality of staircases corresponding to the gate layers are formed in the staircase region of the stack structure, each of the gate layers includes a first gate part at the corresponding staircase and a second gate part connected to the first gate part, and a thickness of the first gate part along the stack direction is less than that of the second gate part along the stack direction; and forming, in the staircase region, a first conductive pillar penetrating through the stack structure along the stack direction, the first conductive pillar being electrically connected with the first gate part.
11. The manufacturing method of the semiconductor device according to claim 10, wherein the step of forming the stack structure further comprises: providing a substrate; forming the plurality of interlayer insulating layers and gate sacrificial layers laminated along the stack direction on the substrate, wherein the plurality of interlayer insulating layers and the plurality of gate sacrificial layers form a plurality of initial staircases in the staircase region, and each of the initial staircases includes one of the gate sacrificial layers and one of the interlayer insulating layers, wherein the one of the gate sacrificial layers and the one of the interlayer insulating layers are in the staircase region and laminated; thinning the gate sacrificial layer at each of the initial staircases; removing the gate sacrificial layer to form a plurality of slots; and forming the plurality of gate layers in the plurality of slots to obtain the stack structure.
12. The manufacturing method of the semiconductor device according to claim 11, wherein, after the step of thinning the gate sacrificial layer at each of the initial staircases, the manufacturing method further comprises: forming a first contact hole in the staircase region, the first contact hole penetrating through the stack structure along the stack direction; and forming an intermediate column in the first contact hole.
13. The manufacturing method of the semiconductor device according to claim 12, wherein the step of forming the plurality of gate layers in the plurality of slots to obtain the stack structure further comprises: removing the intermediate column in the first contact hole; and making the first contact hole penetrate through the plurality of gate layers along the stack direction, etching the first gate part of one of the gate layers in the first contact hole to form a first groove, and etching the second gate part of the other gate layers in the first contact hole to form a second groove.

14. The manufacturing method of the semiconductor device according to claim 13, wherein the step of forming, in the staircase region, a first conductive pillar penetrating through the stack structure along the stack direction further comprises: forming a conductive functional material layer in the first contact hole, wherein the conductive functional material layer is filled in the first groove and covers an inner wall of the second groove; removing at least the conductive functional material layer outside the first groove to form a conducting part in the first groove, wherein the first gate part is around the conducting part and in contact with the conducting part; forming an insulating part in the second groove, wherein the second gate part is around the insulating part; and forming the first conductive pillar corresponding to the first contact hole, wherein the first conductive pillar is electrically connected with the first gate part through the conducting part in the first contact hole.
15. The manufacturing method of the semiconductor device according to claim 14, wherein the step of forming a conducting part in the first groove includes forming a conductive sub-part and a first adhesive buffer sub-part, and wherein the first adhesive buffer sub-part is formed between the first gate part and the conductive sub-part, and the conductive sub-part is formed between the first conductive pillar and the first adhesive buffer sub-part.
16. The manufacturing method of the semiconductor device according to claim 15, wherein the step of forming a plurality of the first conductive pillars corresponding to the respective first contact holes includes forming a conductive body and a second adhesive buffer sub-part covering the conductive body, and wherein the second adhesive buffer sub-part is formed between the conductive sub-part and the conductive body.
17. The manufacturing method of the semiconductor device according to claim 14, wherein: the semiconductor device further comprises a peripheral region on a side of the staircase region away from the array region and a dielectric layer formed at least in the peripheral region, and the step of forming a first contact hole in the staircase region further comprises: using the same process to form the first contact hole, a second contact hole in the staircase region and form a third contact hole in the peripheral region, wherein the first contact hole and the second contact hole both penetrate through the stack structure along the stack direction, and the third contact hole penetrates through the dielectric layer along the stack direction.
18. The manufacturing method of the semiconductor device according to claim 17, wherein the step of forming an intermediate column in the first contact hole further comprises: forming a support pillar in the second contact hole, and the step of forming the first conductive pillar corresponding to the first contact hole further comprises: forming a second conductive pillar in the peripheral region, wherein the second conductive pillar is formed in the third contact hole.
19. A storage system, comprising: a memory including a semiconductor device having a stack structure including a plurality of interlayer insulating layers and gate layers arranged alternately along a stack direction, wherein the stack structure includes an array region and a staircase region adjoining the array region, the stack structure has a plurality of staircases in the staircase region arranged corresponding to the gate layers, each of the gate layers includes a first gate part at the corresponding staircase and a second gate part connected to the first gate part, and a thickness of the first gate part along the stack direction is less than that of the second gate part along the stack direction; a first conductive pillar arranged in the staircase region and penetrating through the stack structure along the stack direction, wherein the first conductive pillar is electrically connected with the first gate part; and a controller which is coupled to the memory and used to control the memory to store data.